Dr. D. Y. Patil Educational Federation's
Dr. D. Y. Patil College of Engineering & Innovation
Approved by AICTE, Recognized by Govt. of Maharashtra,
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
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Department of Computer Engineering

Program name :- Computer Engineering **Class:** S.Y. B. Tech **Div:-** C

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SMARTBRIDGE | SKILLWALLET

FinWise: Personal Finance Advisor Bot

Project Description :-

FinWise (Personal Finance Advisor Bot) is an AI-powered personal finance planning platform designed to help individuals record income, track daily and weekly expenses, and receive a clear, structured budget in Indian Rupees. The platform helps salaried professionals, students, freelancers and household managers understand their spending, savings and financial goals without requiring advanced financial knowledge.

The system combines Flask, SQLAlchemy and an AI engine to automatically analyse income and expense data, generate a personalised 50/30/20 budget plan and highlight overspending categories. Its AI Advisor feature allows users to ask any personal-finance question in natural language — covering SIPs, mutual funds, tax saving, insurance and loans — and receive a clear, conversational answer.

The application is designed with a responsive, modern FinTech interface spread across multiple pages — Dashboard, Transactions, Budget, Goals, Reports and the AI Advisor — and supports both web and mobile users. API keys are securely managed through environment variables and the platform is structured to provide fast, reliable and user-friendly access to personal financial information.

Technology & AI :-

Frontend: React / HTML / CSS / Tailwind / Bootstrap
Backend: Flask / Node.js / Express

AI: Anthropic Claude API, SQLAlchemy-driven financial analysis engine
Database: SQLAlchemy (SQLite/PostgreSQL)
Build Tooling: Vite
Deployment: Firebase Hosting / Render
Security: Environment variables for API keys

Key Features :-

- AI-generated personalised budget plans (50/30/20 rule)
- Income and expense logging by category, in Indian Rupees
- AI Advisor for any financial question — SIPs, mutual funds, stocks, FDs, tax, insurance and loans
- Spending insights that flag overspending categories
- Savings goals with progress tracking
- Structured monthly financial reports and summaries
- Responsive, multi-page interface across 6-7 pages
- Dynamic UI with animations and motion effects
- New user account and unique ID generated for every signup, with no pre-loaded data

Scenarios :-

Scenario 1: A salaried professional logs monthly income and daily expenses across categories such as rent, food, transport and entertainment. The system analyses the spending data, identifies overspending areas, and generates a personalised budget plan with actionable saving suggestions.

Scenario 2: A college student uses FinWise to manage a limited monthly allowance across essential and discretionary spending categories. The platform helps the student set realistic budget limits, track spending habits, and receive AI-powered recommendations for reducing unnecessary expenditure.

Scenario 3: A freelancer with variable monthly income uses FinWise to log earnings from multiple clients and track project-related expenses. The system adapts budget

recommendations to fluctuating income, highlights months with reduced saving capacity, and suggests strategies for building an emergency fund.

Scenario 4: A household manager uses FinWise to monitor shared family income and expenses across categories such as groceries, utilities, education and healthcare. The bot generates a consolidated monthly financial report, identifies budget overruns, and suggests tailored saving strategies to meet family financial goals.

Milestones :-

Milestone 1 – Project Planning & Requirement Analysis

Define the objective of FinWise, identify target users (salaried professionals, students, freelancers, household managers), finalize core features, and design the overall application workflow.

Milestone 2 – Backend & Data Model Setup

Build the Flask and SQLAlchemy backend, design the database schema for users, income, expenses and categories, and implement APIs for logging and retrieving transactions.

Milestone 3 – AI Budget Generation & Insights

Integrate the AI engine to analyse spending data, generate the 50/30/20 budget plan, and produce personalised saving suggestions and spending insights.

Milestone 4 – AI Advisor & Frontend Build

Develop the AI Advisor chat feature for natural-language financial questions, and build the multi-page React/Tailwind/Bootstrap frontend with dynamic animations and effects.

Milestone 5 – Testing, Optimization & Deployment

Test the application across mobile and desktop devices, fix API and UI issues, improve performance and reliability, secure environment variables, and deploy the final application.

Conclusion :-

FinWise provides a simple and interactive way to take control of personal finances. By combining structured income and expense tracking, AI-driven budget generation and an AI Advisor for financial questions, the platform helps users understand their spending, savings and financial goals. The project demonstrates how AI and modern full-stack web technologies can be used to make personal finance management more accessible, disciplined and useful for everyday individuals.

Expected Outcome :-

FinWise aims to make personal finance management more accessible by reducing the manual complexity of budgeting and presenting income, expenses and savings in a simple, structured and understandable format. The platform combines automated budget generation, AI-powered financial guidance and structured monthly reporting to create an interactive personal finance planning experience.